Unit-5
Environmental Pollution Regulations & Act

Topics to be covered: Environmental Protection Act 1986, Initiatives by Non- governmental organizations (NGO's), Population growth explosion- family welfare program, Environment and human health, Population growth, Environmental Education, Women Education.

Environmental Protection Act 1986:

The Act came into force on November 19, 1986, the birth anniversary of our Late Prime Minister Indira Gandhi, who was a pioneer of environmental protection issues in our country. The Act extends to whole of India. The Constitution of India clearly states that it is the duty of the state to 'protect and improve the environment and to safeguard the forests and wildlife of the country'.
As compared to all other previous laws on environment protection, the Environment (Protection) Act, 1986 is a more effective and comprehensive measure to fight the problem of pollution.

Objects:

i. The Environment Protection Act is a means to implement the decisions of the UN Conference on the Human Environment held in Stockholm (June 1972).
ii. The Environment Protection Act also seeks to enact a general blanket law on environmental protection, dealing with all aspects of pollution and harm to the nature and not limiting its scope to just one type of pollution or pollutants.
iii. The exhaustive nature of Environment Protection Act also ensures that no ambit of environmental protection is left and all hazards' to the environment are absolutely roofed and addressed under the Act.
iv. The Act also provides punishment (deterrent in nature) to those responsible for causing harm to the environment or endangering it.
v. The Act provides for a scheme and a mechanism of working of various already existing regulatory authorities and also creates more agencies for furtherance of environment protection.
vi. The Act also aims at promoting sustainable development as a means to achieve the end of prosperity and opulence.

THE AIR (PREVENTION AND CONTROL OF POLLUTION) ACT: The Government passed this Act in 1981 to clean up our air by controlling pollution. Sources of air pollution such as industry, vehicles, power plants, etc. are not permitted to release particulate matter, lead, carbon monoxide, sulphur dioxide, nitrogen oxide, volatile organic compounds (VOCs) or other toxic substances beyond a prescribed level. To ensure this, Pollution Control Boards (PCBs) have been set up by Government to measure pollution levels in the atmosphere and at certain sources by testing the air. This is measured in parts per million or in milligrams or micrograms per cubic meter.
The main objectives of the Act are as follows:
(a) To provide for the Prevention, Control and abatement of air pollution.

(b) To provide for the establishment of Central and State Boards with a view to implement the Act.
(c) To confer on the Boards the powers to implement the provisions of the Act and assign to the Boards functions relating to pollution.

THE WATER (PREVENTION AND CONTROL OF POLLUTION) ACT: The Government has formulated this Act in 1974 to be able to prevent pollution of water by industrial, agricultural and household wastewater that can contaminate our water sources. The main objectives of the Water Act are to provide for prevention, control and abatement of water pollution and the maintenance or restoration of the wholesomeness of water. It is designed to assess pollution levels and punish polluters. The Central Government and State Governments have set up Pollution Control Boards that monitor water pollution.

THE WILDLIFE PROTECTION ACT: This Act passed in 1972, deals with the declaration of National Parks and Wildlife Sanctuaries and their notification. It establishes the structure of the State's wildlife management and the posts designated for Wildlife Management. It provides for setting up Wildlife Advisory Boards.
The Amendment to the Wildlife Protection Act in 2002 is more stringent and prevents the commercial use of resources by local people. It has brought in new concepts such as the creation of Community Reserves. It has also altered several definitions. For instance in animals, fish are now included. Forest produce has been redefined to ensure protection of ecosystems. While there are several changes, the new Act still has serious issues concerned with its implementation. Laws are only as good as the ones that can be complied with.

INITIATIVES BY NON- GOVERNMENTAL ORGANIZATIONS (NGO'S)

Greenpeace. International NGO with a goal to "ensure the ability of the Earth to nurture life in all its diversity."

Earth Institute Center for Environmental Research and Conservation. Sustainability think tank based at Columbia University.

Earth Island Institute. Founded by David R. Brower, acts as an environmental umbrella organization, providing people and groups the capability to develop initiatives.

Earth Justice. Environmental interest organization headquartered in San Francisco dedicated to litigating environmental issues.

Environmental Defense Fund. Works on global warming, ecosystem restoration, oceans, and human health. See their list of publications.

Fauna and Flora International. Works to conserve threatened species and ecosystems worldwide into account human needs.

Nature Friends International. Global environmental umbrella organization.

Global Footprint Network. International think tank that provides ecological footprint accounting tools to drive policy decisions in a resource-constrained world.

International Union for Conservation of Nature. Union composed of government and civil society organizations to provide the knowledge and tools that enable human progress, economic development and nature conservation to take place together.

Nature Conservancy. Conservation organization working to protect ecologically important lands and waters for nature and people.

Natural Resources Defense Council. International environmental advocacy group with a staff of hundreds of lawyers, scientists and policy experts.

Wetlands International. Global not-for-profit organization dedicated to the conservation and restoration of wetlands.

World Agroforestry Centre. Generates knowledge about the benefits of agroforestry to develop policy and practices for improved livelihoods and environmental benefits.

World Wildlife Fund. International Conservation Organization - see also the International Secretariat based in Geneva.

POPULATION EXPLOSION – FAMILY WELFARE PROGRAM

The world population is growing by more than 90 million per year, of which 93% is in developing countries. This will essentially prevent their further economic 'development'. In the past, population growth was a gradual phenomenon and the Earth's ability to replenish resources was capable of adjusting to this increase. In the recent past, the escalation in growth of human numbers has become a major cause of our environmental problems.

The increasing pressures on resources place great demands on the in-built buffering action of nature that has a certain ability to maintain a balance in our environment. However, current development strategies that essentially lead to short-term gains have led to a breakdown of our Earth's ability to replenish the resources on which we depend.

In response to our phenomenal population growth, India seriously took up an effective Family Planning Program which was renamed the Family Welfare Program. Slogans such as 'Hum do hamare do' indicated that each family should not have more than two children. It however has taken several decades to become effective.

At the global level by the year 2000, 600 million, or 57% of women in the reproductive age group, were using some method of contraception. However the use of contraceptive measures is higher in developed countries – 68%, and lowers in developing countries - 55%. Female sterilization is the most popular method of contraception used in developing countries at present. This is followed by the use of oral contraceptive pills and, intrauterine devices for women, and the use of condoms for men. India and China have been using permanent sterilization more effectively than many other countries in the developing world.

Informing the public about the various contraceptive measures that are available is of primary importance. This must be done actively by Government Agencies such as Health and Family Welfare, as well as Education and Extension workers. It is of great importance for policy makers and elected representatives of the people – Ministers, MPs, MLAs at Central and State levels – to understand the great and urgent need to support Family Welfare.

The media must keep people informed about the need to limit family size and the ill effects of a growing population on the world's resources. The decision to limit family size depends on a couple's background and education. This is related to Government Policy, the effectiveness of Family Welfare Programs, the educational level, and information levels in mass communication.

Free access to Family Welfare information provided through the Health Care System, is in some cases unfortunately counteracted by cultural attitudes. Frequently misinformation and inadequate information are reasons why a family does not go in for limiting its size.

The greatest challenge the world now faces is how to supply its exploding human population with the resources it needs. It is evident that without controlling human numbers, the Earth's resources will be rapidly exhausted. In addition economically advanced countries and rich people in poorer countries use up more resources than they need.

The first 'green revolution' in the '60s produced a large amount of food but has led to several environmental problems. Now, a new green revolution is needed, to provide enough food for our growing population, that will not damage land, kill rivers by building large dams, or spread at the cost of critically important forests, grasslands and wetlands.

Human populations will inevitably expand from farm lands into the remaining adjacent forests.

Many such encroachments in India have been regularised over the last few decades. But forest loss has long-term negative effects on water and air quality and the loss of biodiversity is still not generally seen as a major deterrent to human well-being. The extinction of plant and animal species resulting from shrinking habitats threatens to destroy the Earth's living web of life.

Energy use is growing, both due to an increasing population, and a more energy hungry lifestyle that increasingly uses consumer goods that require large amounts of energy for their production, packaging, and transport. Our growing population also adds to the enormous amount of waste.

With all these linkages between population growth and the environment, Family Welfare Programs have become critical to human existence.

ENVIRONMENT AND HUMAN HEALTH

- Living entities depend upon their environment for energy requirements.
- Some of these requirements are pure water, clean air, unadulterated and nutritious food, a disease-free community to live in, etc.
- These factors are known to add to the longevity of human history.
- It is also a known fact that sanitation, agriculture, treated water, and personal and community hygiene have an impact on human health.
- One cannot overlook the fact that apart from supporting and nurturing human life, it can also cause diseases.
- One of the most common causes of death rate is the lack of necessities.
- Environmental risks can increase the chances of contracting heart diseases, cancers, and various other illnesses.
- Untreated drinking water, poor hygiene, and improper sanitation cause infectious diseases such as cholera, diarrhea, dengue, etc.

Impacts of the environment on human health:

- The environment relates to all the living and non-living things surrounding us.
- The environment has a great impact on human health.
- We interact with the environment daily.
- It is very much essential to keep our environment healthy to protect our lives from different environmental hazards.

- The environmental hazards are physical, chemical, and biological hazards.
- Examples of physical hazards are airborne particles, humidity, equipment design, radiations, etc.
- Examples of biological hazards are viruses, microbial agents, insects, rodents, animals and plants, etc.
- Examples of chemical hazards are pesticides, insecticides, herbicides, lead, acids, chlorine, and other caustic substances.
- There are a minute amount of chemicals and toxins in the daily food we eat, the breathing air, and our drinking water.
- The effects of these environmental hazards include cancers, respiratory system disorders asthma, allergies, and other illnesses.
- Many infectious diseases in this environment are all carried by pathogens.

ENVIRONMENTAL EDUCATION

It is now widely acknowledged that the impact of human activities on environment is significant and will have serious consequences for future generations. Environment education has a significant role in making the youth conscious about their environment, adopting green social behaviour and thus responding to the environment crises. Recognizing the importance and need of environment education, India has initiated several efforts including making the environment education compulsory at all level of education. Today when environmental conditions are changing adversely and all living beings are suffering from the negative consequences of environmental pollution and climate changes, there is a need to reorient the curriculum of environment education for making it more attractive and responsive to the local environmental issues. Moreover, related policies are also needed to be restructured for sensitizing the people, especially school and college level students about the issues related to environment management. The components of environmental education are:

- Awareness and sensitivity to the environment and environmental challenges
- Knowledge and understanding of the environment and environmental challenges
- Attitudes of concern for the environment and motivation to improve or maintain environmental quality
- Skills to identify and help resolve environmental challenges
- Participation in activities that lead to the resolution of environmental challenges

WOMEN EDUCATION

Women are the building blocks of society. They play a vital role in empowering every aspect of society and the nation. A nation is considered as well established and empowered only when the female population of the country is established, empowered, and safe. Education is that key element that helps every individual conquer the greatest war and also create the most impactful changes in the world.

A woman is such an individual who has the knowledge and temperament to create lives with meaning. Promoting women's education is as necessary and useful as oxygen is for breathing.

India has been able to generate some of the famous leaders who are female. Yet, in most of the portions of the country, women still lack the benefit of getting an education and are deprived of their basic amenities and rights.

Importance of women education:

1. Elimination of crime against women: Education plays a vital weapon in eliminating manage social crimes and evils against women prevailing in society.. Social customs such as Sati, female infanticide, dowry , class trade and other harmful customary practice can be eradicated through female education. An educated women stands up for the injustice patted against women in the family or society against other girls or women .She is a pivotal in a civilized society and influences the beliefs and thoughts of its members.
2. self-Reliance: Education is vital for women as it makes women become self-reliant and eliminates the need to depend on a third person for her and her family survival. She becomes aware of a rights and Employment on an equal plank with men and finds the needs of a family. Financially independent woman raise her voice against the prevailing old social customs and injustice.

3. Improvised standard of living: Female education improvisers and elevates the standard of living. A family relying on double wages leads to a more satisfied and happy family in comparison to a family that relies on a single parent income. Two incomes under the same roof improve the quality of living and ensure to facilitate the importance of female education in the family and Society. An educated woman equally pays as the male member and adds and elevates the family's financial needs and the standard of family.
4. Prevent social exclusion: An uneducated woman is likely to work as domestic help or in extreme cases become a victim of flesh trade over the opposite gender. Women often get secluded from society who spends their life as domestic help or any other manual jobs. The seclusion or exclusion of women by society leads to Physical as well as psychological taumas ailments. An educated women brews balance the society.